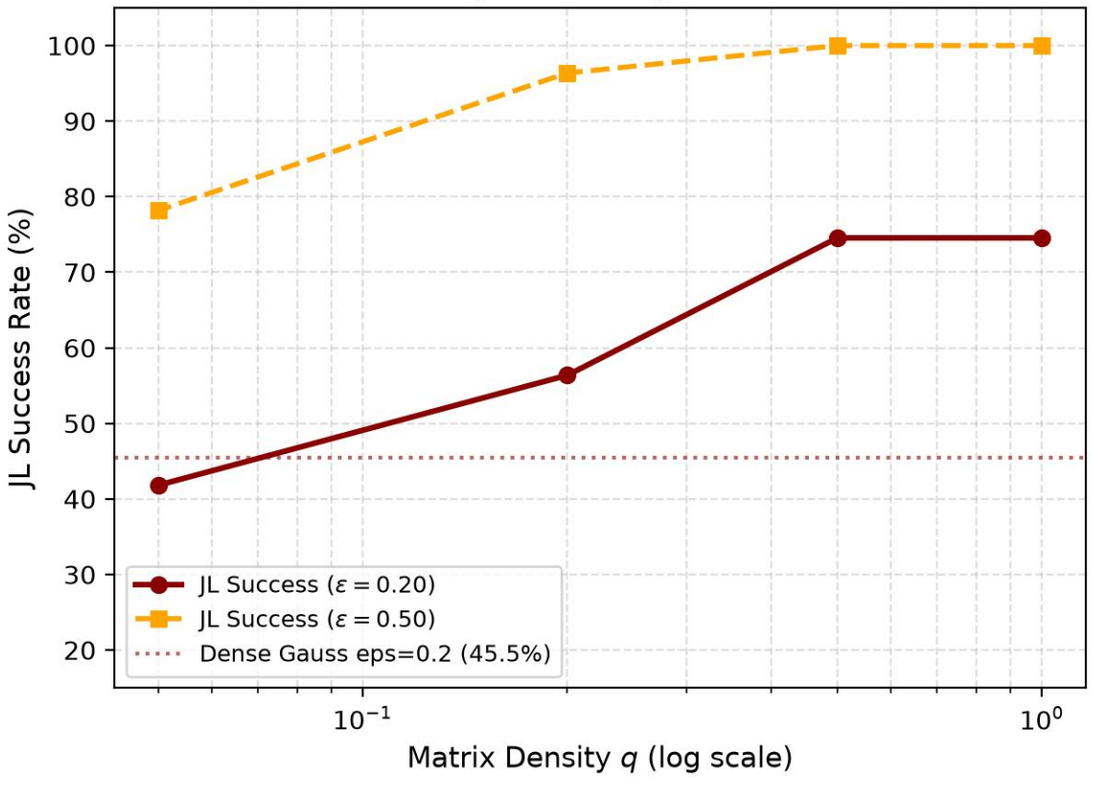
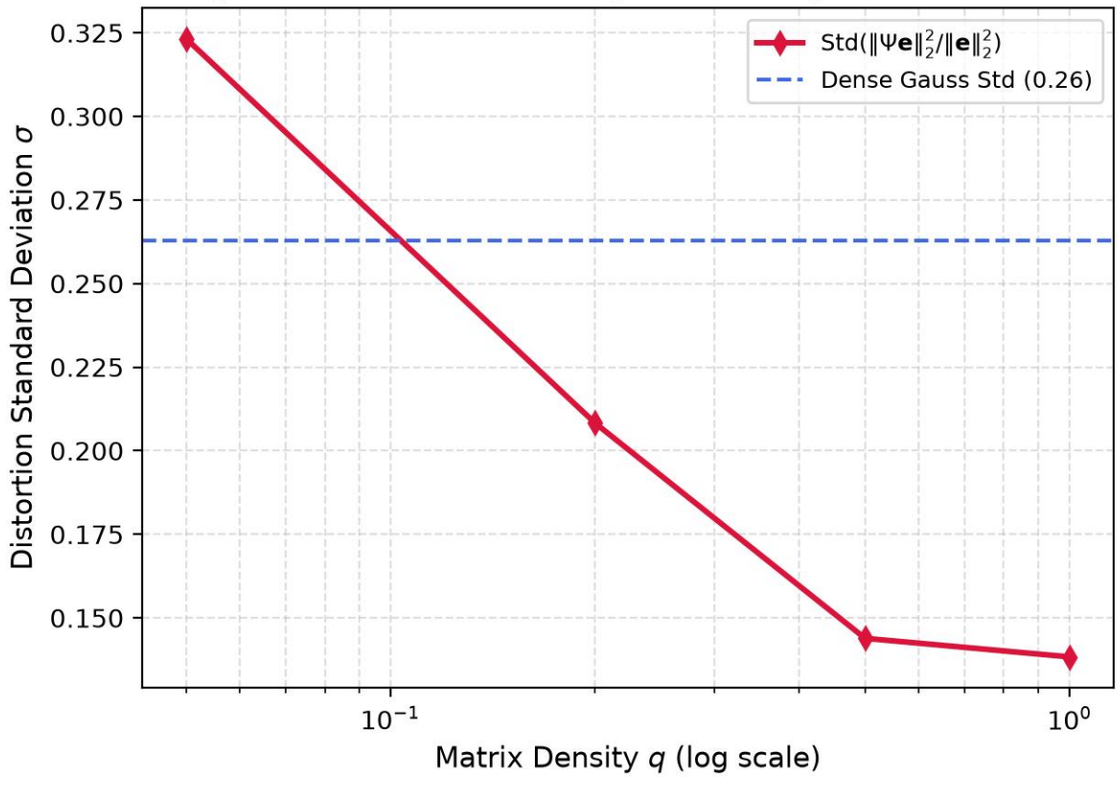
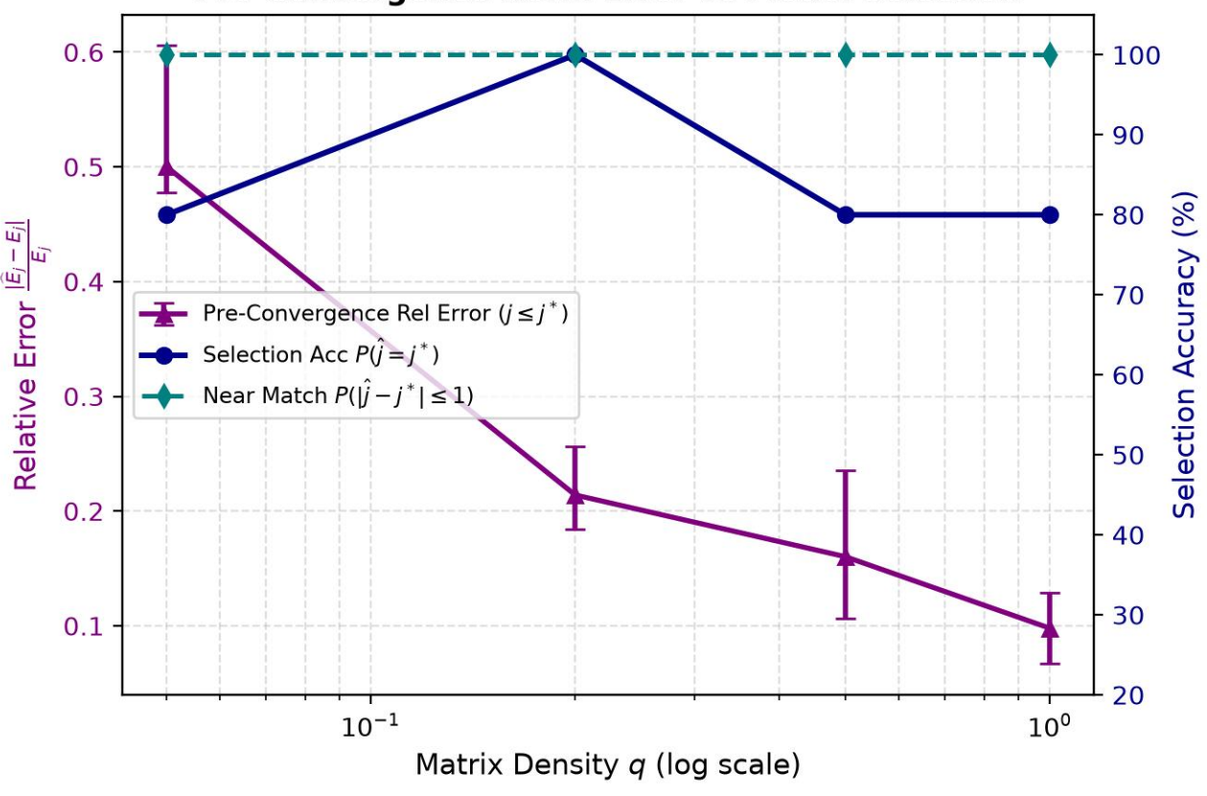
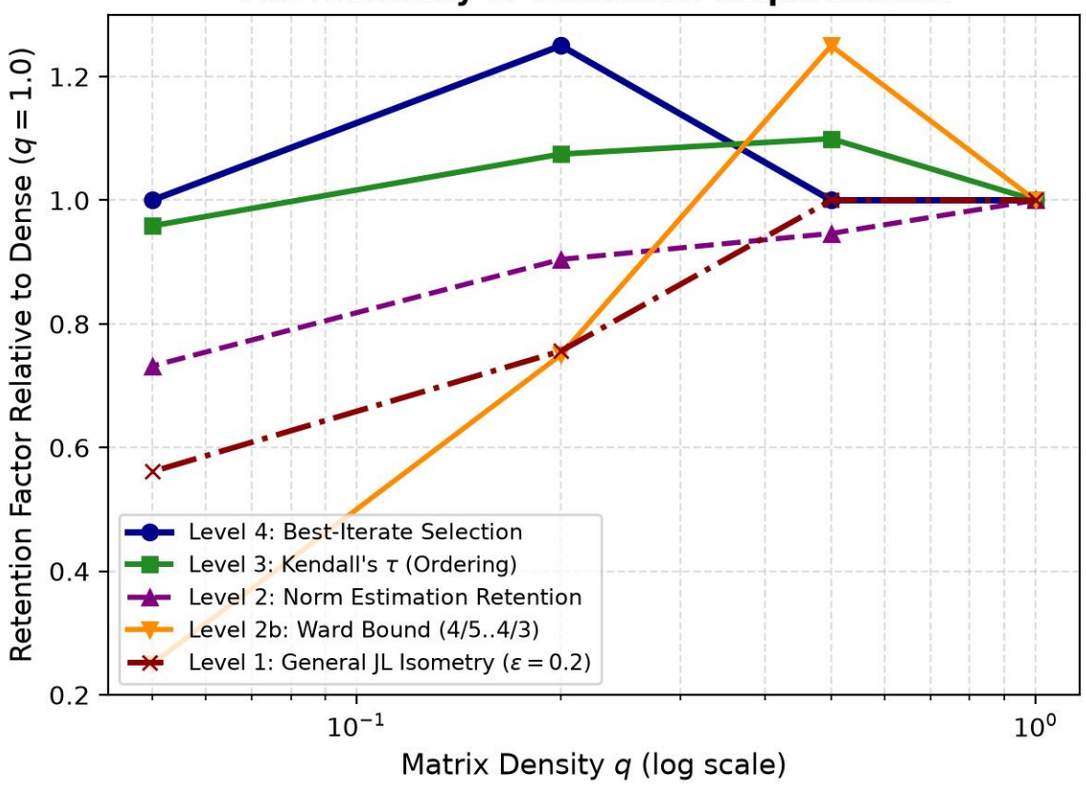

Level 1: General JL Isometry $(1 \pm \varepsilon)$ Preservation
JL Success Rate (%)
Matrix Density q (log scale)
JL Success ($\varepsilon = 0.20$)
JL Success ($\varepsilon = 0.50$)
Dense Gauss eps=0.2 (45.5%)


Projection Tail Variance Expansion (Higher = Unstable)
Distortion Standard Deviation σ
Matrix Density q (log scale)
Std($\|\Psi\mathbf{e}\|_2^2/\|\mathbf{e}\|_2^2$)
Dense Gauss Std (0.26)


Pre-Convergence Norm Error vs Model Selection
Relative Error $\frac{|\hat{E}_j - E_j|}{E_j}$
Selection Accuracy (%)
Matrix Density q (log scale)
Pre-Convergence Rel Error ($j \le j^*$)
Selection Acc $P(\hat{j} = j^*)$
Near Match $P(|\hat{j} - j^*| \le 1)$


The Hierarchy of Validation Requirements
Retention Factor Relative to Dense ($q = 1.0$)
Matrix Density q (log scale)
Level 4: Best-Iterate Selection
Level 3: Kendall's τ (Ordering)
Level 2: Norm Estimation Retention
Level 2b: Ward Bound (4/5..4/3)
Level 1: General JL Isometry ($\varepsilon = 0.2$)